

PrintDeck

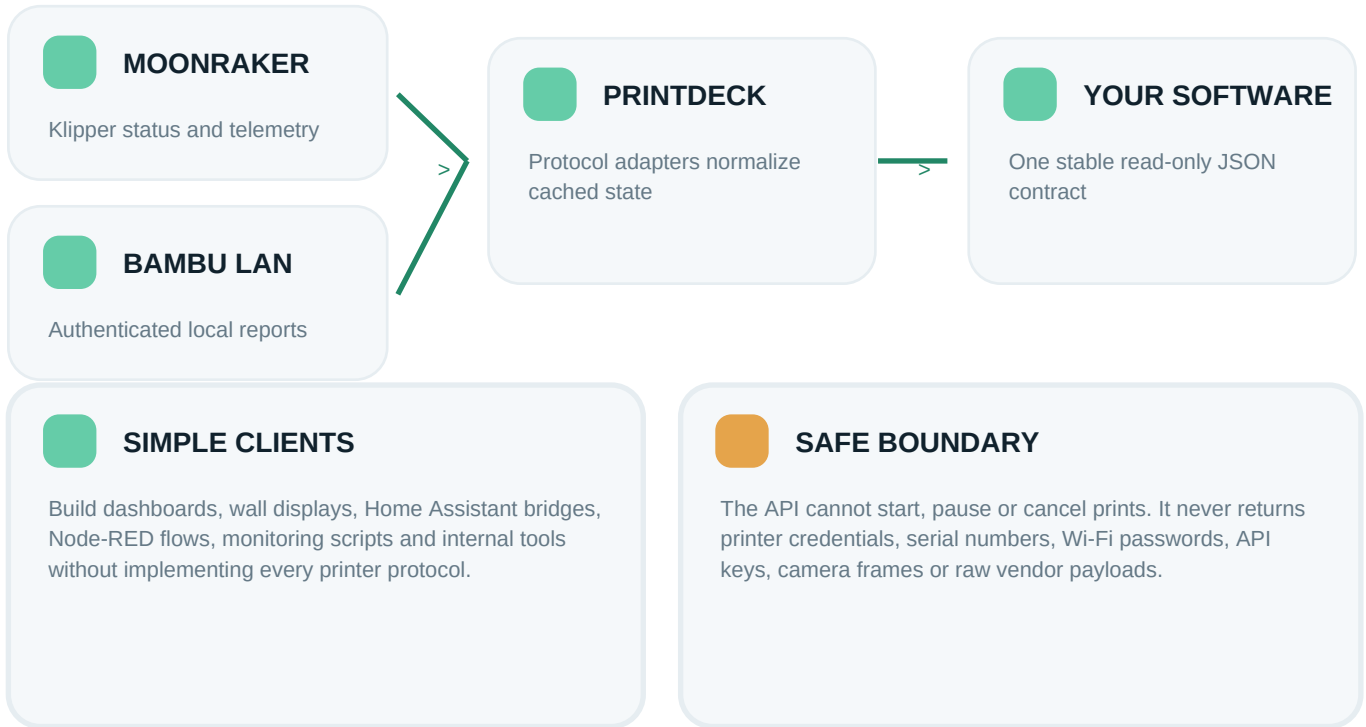
UNIFIED PRINTER API

One local, read-only JSON format for every printer



One API between your software and every printer

Printer vendors and local print servers expose different protocols, field names and status models. PrintDeck already translates those inputs for its screens. The Unified Printer API exposes that normalized, cached state as versioned JSON.



DATA PATH

```
printer formats -> PrintDeck normalized cache -> GET /v1/... -> JSON
```

No webhooks, no cloud account, no remote relay and no control endpoints.

Enable access and authenticate

01

Open Web Config

Open the local address shown by PrintDeck, choose Device Settings and select Unified Printer API above Backup & Restore.

02

Enable the API

Choose Enable API. PrintDeck creates a random token and stores it locally on the device.

03

Copy the token

Use Copy token and store it as a secret in your application. Do not confuse it with a Bambu Lab LAN access code.

04

Send Bearer auth

Add the token to every request in the Authorization header. A missing or invalid token returns HTTP 401.

FIRST REQUEST

```
curl http://PRINTDECK-IP/v1/printers/status \  
-H 'Authorization: Bearer pd_your_token_here'
```

TRUSTED LAN ONLY

PrintDeck serves its local device API over HTTP. Keep it on a trusted network, do not forward the port to the internet and regenerate the token if it may have been exposed.

Endpoint reference

GET /v1/info

PrintDeck, hardware, firmware and safe network details.

GET /v1/printers

Saved printer identities, addresses and capabilities.

GET /v1/printers/status

Compact status array for every saved printer.

GET /v1/printers/{id}

One printer identity and normalized capabilities.

GET /v1/printers/{id}/status

Connection, job, timing and temperature data.

GET /v1/printers/{id}/nozzles

Tool, material, filament and nozzle temperatures.

GET /v1/printers/{id}/materials

AMS, AMS Lite and external spool data when known.

All routes are read-only. Query strings are not required. Printer IDs remain stable until a profile is deleted and recreated.

Unavailable fields are null. Known offline printers still return HTTP 200 with `stale: true`. Unknown printer IDs return HTTP 404.

PrintDeck and printer identity

DEVICE INFORMATION

```
GET /v1/info
{
  "api_version": "v1",
  "product": "PrintDeck",
  "firmware_version": "1.0.3",
  "hardware": "amoled_1_75",
  "network": {
    "wifi_name": "Workshop Wi-Fi",
    "ipv4": "192.168.1.42",
    "hostname": "printdeck.local"
  },
  "read_only": true
}
```

PRINTER LIST

```
GET /v1/printers
{
  "api_version": "v1",
  "printers": [{
    "id": 1,
    "name": "Workshop Voron",
    "protocol": "moonraker",
    "manufacturer": "Voron",
    "model": "2.4",
    "endpoint": "192.168.1.50:7125",
    "selected": true,
    "reachability": "online",
    "capabilities": {"status": true, "nozzles": true, "materials": true}
  }]
}
```

Status and freshness

SELECTED PRINTER - FULL DETAIL

```
GET /v1/printers/1/status
{
  "api_version": "v1",
  "status": {
    "printer_id": 1,
    "connection": {
      "state": "online",
      "reachability": "online",
      "detail_level": "full",
      "stale": false,
      "updated_at_ms": 418520
    },
    "job": {
      "phase": "printing", "kind": "print",
      "activity": "printing", "name": "Bracket",
      "progress_percent": 42.5,
      "elapsed_seconds": 1260, "remaining_seconds": 1740,
      "current_layer": 48, "total_layers": 112
    },
    "temperatures": {
      "nozzle_current_c": 200.4, "nozzle_target_c": 200,
      "bed_current_c": 59.8, "bed_target_c": 60,
      "chamber_current_c": null
    }
  }
}
```



DETAIL LEVEL

full means the selected printer has live normalized telemetry. summary means PrintDeck has only the bounded printer-list probe.



STALE DATA

Always inspect stale and connection.state. Cached data can remain useful while a printer reconnects or is offline.

Nozzles and materials

TOOLHEADS

```
GET /v1/printers/1/nozzles
{
  "printer_id": 1, "detail_level": "full", "stale": false,
  "nozzles": [{
    "id": "T0", "active": true, "state": "ready",
    "diameter_mm": 0.4,
    "temperature": {"current_c": 133, "target_c": 200},
    "material": {"type": "PLA", "color": "#111111FF"},
    "filament_detected": true
  }]
}
```

MATERIAL SYSTEM

```
GET /v1/printers/2/materials
{
  "printer_id": 2, "available": true,
  "system": "ams_or_ams_lite", "stale": false,
  "slots": [{
    "id": "slot-0", "installed": true, "feeding": true,
    "material": "PLA", "color": "#E6B422FF",
    "remaining_percent": 72,
    "source_unit": 0, "source_slot": 0
  }],
  "external_spools": []
}
```

Array order follows the normalized tool or feeder order reported by PrintDeck. Empty arrays are valid and do not indicate an API error.

Errors, integration rules and privacy



HTTP 401

The Authorization header is missing or the Bearer token is invalid. Copy the current token from Web Config.



HTTP 403

The Unified Printer API is disabled. Enable it locally in Web Config before retrying.



HTTP 404

The route is unknown or the printer ID no longer exists. Refresh GET /v1/printers.



HTTP 429

The client exceeded one authorized request per second. Wait for Retry-After: 1 before retrying.

Client checklist

- Keep the token outside source control and logs.
- Use the printer ID as the stable key, not its display name.
- Handle null fields and empty arrays without inventing values.
- Read connection.state, detail_level and stale before telemetry.
- Send at most one request per second; faster requests return HTTP 429.
- Do not expose the PrintDeck port or token outside the trusted local network.

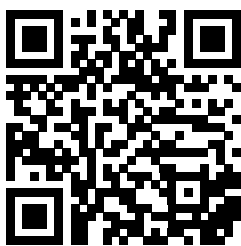
RATE-LIMIT RESPONSE

```
HTTP/1.1 429 Too Many Requests
Retry-After: 1
{"error":{"code":"rate_limited","message":"Unified Printer API allows one request per second."}}
```


PrintDeck

ONE FORMAT. MANY PRINTERS.

Build local software against the data model PrintDeck already uses for its own screens.



DEVELOPER DOCUMENTATION

Examples, use cases and current guide

Scan to open the Unified Printer API page and download the latest documentation.

<https://printdeck.xyz/unified-printer-api/>